

Addendum: A Galois-Fluid Stress Test of the $\chi(3)$ Channel

Frobenius traces $a_\ell(E_p)$ confirm CRT orthogonality

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The $\chi(3)$ -channel principle predicts that the 6N wing ($p \bmod 3$) can reach an L -function-derived invariant of E_p only through the $q = 3$ Euler factor. This addendum subjects that prediction to a deliberately adversarial test, using a quantity drawn from a *different* Euler factor: the Frobenius trace $a_\ell(E_p)$ for $\ell = 7, 13$. If the wing could influence the curve’s local data at $\ell \neq 3$, the principle would have a leak. It does not.

Design. For every prime $3 < p < 2 \times 10^4$ (2260 primes) we form $E_p : y^2 = x^3 - p$ and extract $a_7(E_p)$ and $a_{13}(E_p)$ via `E.ap( $\ell$ )` in SageMath. We record, alongside the 6N wing, the residues $p \bmod 7$ and $p \bmod 13$. These residues are not noise to be stripped: they are the **positive control**. The reduction $E_p \bmod \ell$ depends only on $-p \bmod \ell$, so a_ℓ is a deterministic function of $p \bmod \ell$; the control *must* fire if the instrument works. The main test then asks whether the wing adds anything.

Result. The two questions separate as cleanly as an experiment in this series ever has.

	controller	dependence of a_ℓ
Positive control	$p \bmod \ell$ ($\ell = 7, 13$)	$\eta^2 = 1.000$ (fully determined)
Main test	$p \bmod 3$ (the 6N wing)	$ d \approx 0.005$ (null)

The positive control is not approximate: each residue class $p \bmod 7$ maps to a single fixed value of a_7 (and likewise a_{13} across its twelve classes), so $p \bmod \ell$ explains 100% of the variance. The main test is uniformly null — for a_7 , $d = -0.006$; for a_{13} , $d = -0.004$; the same for $|a_\ell|$, a_ℓ^2 , and the Sato–Tate angle $\theta = \arccos(a_\ell/2\sqrt{\ell})$ (all $|d| < 0.008$).

Interpretation: a_ℓ is a CRT coordinate. The result is the cleanest possible display of orthogonality. One and the same quantity a_ℓ is *completely* determined by $p \bmod \ell$ yet *completely* independent of $p \bmod 3$. This is not the wing being blocked by a barrier; it is the Chinese Remainder Theorem itself: $p \bmod 7$ and $p \bmod 3$ are independent coordinates, so a function of one carries no information about the other. The probe was framed as “decrepitating the CRT lattice to extract a deep fluid,” but the fluid it extracts — a_ℓ — *is* a CRT coordinate: bombarding p at temperature ℓ yields exactly $p \bmod \ell$, an axis orthogonal to the wing.

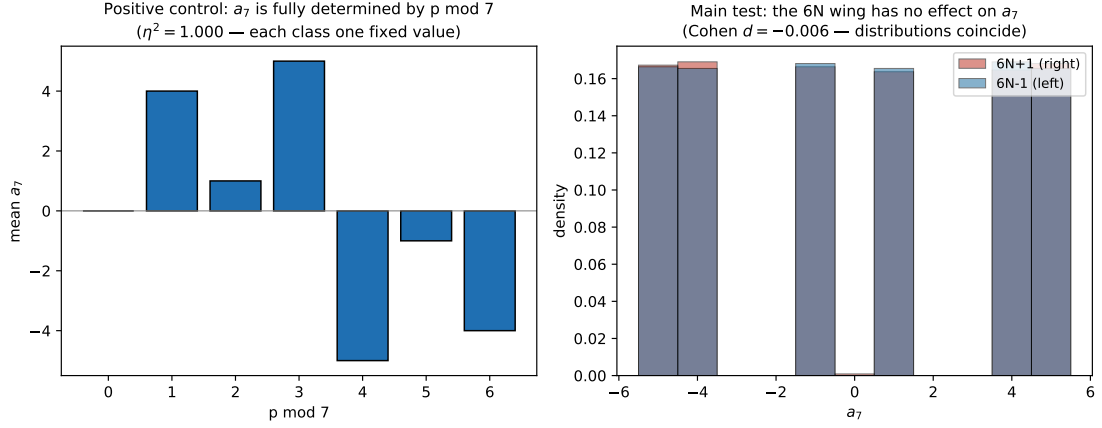


Figure 1: Left: the positive control fires perfectly — a_7 is a fixed function of $p \bmod 7$ ($\eta^2 = 1.000$). Right: the two wings' a_7 distributions coincide (Cohen $d = -0.006$). The same quantity is fully determined by $p \bmod 7$ and entirely independent of $p \bmod 3$.

Consequence for the principle. This independently verifies the $\chi(3)$ -channel principle on its hardest case. The wing acts only in the $q = 3$ factor; a_7 and a_{13} live in the $q = 7$ and $q = 13$ factors; different doors, no leak. It also closes the family: any probe “wing vs a_ℓ ” for any ℓ will return $\eta^2 = 1$ against $p \bmod \ell$ and $d = 0$ against the wing, because CRT makes all the residue axes mutually orthogonal. No further ℓ need be drilled.

Archival note. Data (`Galois.Fluid.Isotopes.Log.csv`, 2260 primes), the extraction script (`Galois_fluid_log.sage`), and the analysis (`Galois_fluid_analysis.py`) are archived with this addendum at <https://github.com/Ruqing1963/6N-chi3-channel>. The positive control ($\eta^2 = 1.000$) certifies the measurement; the result carries the highest confidence of any negative in the series, because the instrument is provably working and the null is exact.